Google Cloud H2S

Gen AI Academy APAC Edition

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Guidelight AI

AI Daily Independence Copilot for Visually Impaired Users

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Problem Statement: Build an AI-powered multi-agent system using Google ADK, MCP & Gemini to solve a real-world accessibility challenge

Google ADK · Gemini 3.1 Flash-Lite · MCP · Accessibility-First

Brief About the Idea

The Problem

🌍 2.2 billion people worldwide live with vision impairment.
Current assistive tools can READ — but cannot REASON, PLAN, or ACT.

❌ Fragmented tools that don't coordinate with each other
❌ No proactive safety monitoring (missed meds, overloaded schedules)
❌ Increased dependence on caregivers for daily planning

Our Solution — Guidelight AI

✅ Multi-agent AI copilot that plans, schedules, tracks & safeguards daily routines
✅ Voice-first, hands-free operation via Web Speech API + TTS
✅ Proactive safety alerts for medication, overload, and risky transitions
✅ Just say what you need — AI agents handle everything else

💡 **"Not a chatbot — an intelligent, multi-agent execution system."**

Solution Approach

How We Used ADK + MCP

5 specialized ADK agents — Planner, Calendar, Tasks, Safety, Coordinator.
MCP tools provide standardized Calendar & Task Store interfaces.
Deterministic sequential pipeline ensures reliable demo execution.

Real-World Problem & Impact

Visually impaired users gain independence through AI that reasons about their entire day — scheduling, reminding, tracking, and protecting.
Reduces caregiver dependence. Increases daily autonomy.

Core Workflow — Optimized to 2 LLM Calls

1. User speaks intent
2. Planner Agent generates structured JSON schedule (LLM Call 1)
3. Calendar & Tasks populated programmatically (no LLM)
4. Safety + Summary Agent evaluates risks & summary (LLM Call 2)
5. Results delivered via accessible tabbed UI with voice readout

Opportunities — USP of the Solution

🏆 What Makes Guidelight AI Different

🛡️ **Safety-First Agent**

No other daily planner proactively detects medication misses, schedule overload, cognitive fatigue, or risky transitions.

🧠 **Reasoning, Not Just Reading**

Existing assistive tools ARE screen readers. Guidelight AI REASONS holistically across the entire day.

⚡ **2-Call Optimized Pipeline**

60% fewer API calls, ~70% fewer tokens. Calendar + Tasks are programmatic.

🎤 **True Voice-First Design**

Full hands-free operation — voice input to spoken summary output.

♿ **Accessibility as Architecture — WCAG, ARIA, keyboard-nav, high-contrast. Day 1.**

"From fragmented tools to an intelligent, coordinated copilot."

Features Offered by the Solution

Multi-Agent Pipeline
5 ADK agents in deterministic sequence — Planner, Calendar, Tasks, Safety, Coordinator

MCP Tool Integration
Calendar & Task Store follow Model Context Protocol with conflict detection

Proactive Safety Agent
Detects overloaded schedules, missed medications, risky transitions & cognitive overload

2-Call Optimization
5 agent calls → 2 LLM calls. Calendar + Tasks are programmatic. ~70% fewer tokens

Voice Input + TTS Output
Web Speech API for hands-free input; SpeechSynthesis reads the summary aloud

Interactive Task Checklist
Mark tasks complete with live progress bar via REST API

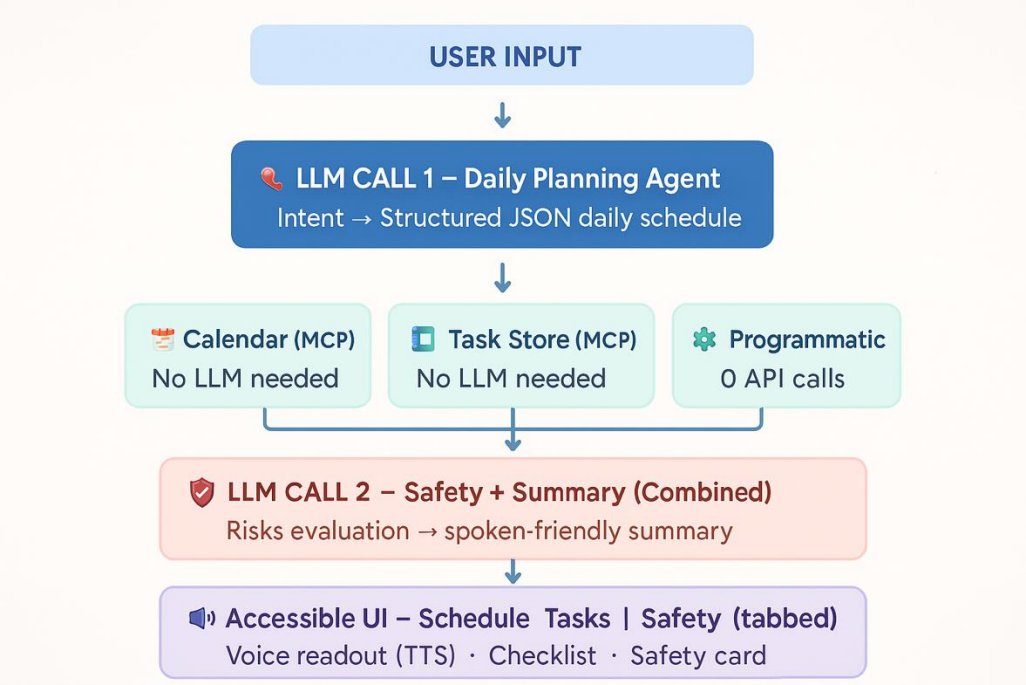
Color-Coded Calendar
Visual schedule with category dots — medication, meals, rest, appointments

6 Quick Scenarios
Medical · Social · Restful · Busy · Navigation · Assistive Tech Day

WCAG Accessible UI
Keyboard-nav, ARIA roles, skip links, high-contrast, screen-reader optimized

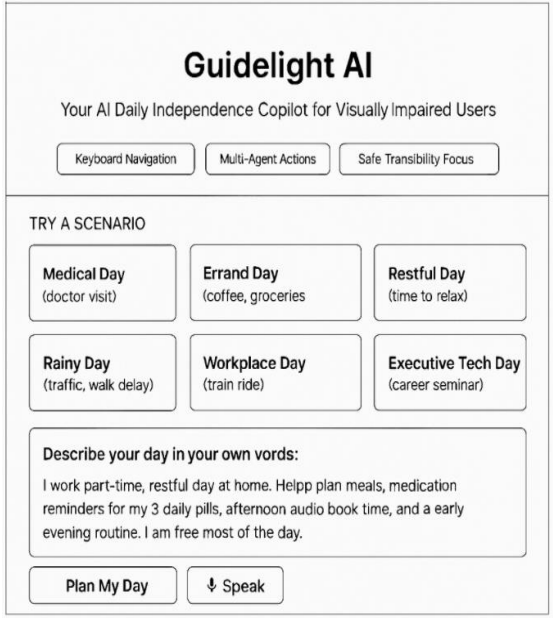
Tabbed Results
Schedule, Tasks & Safety in compact pill-shaped tabs with badge counts

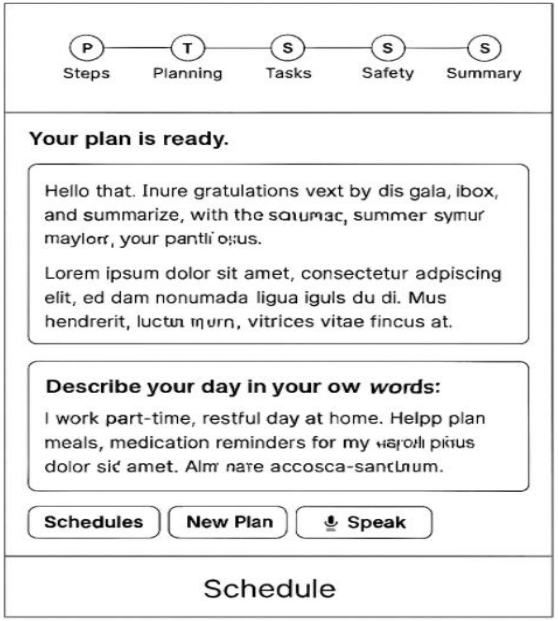
Process Flow Diagram



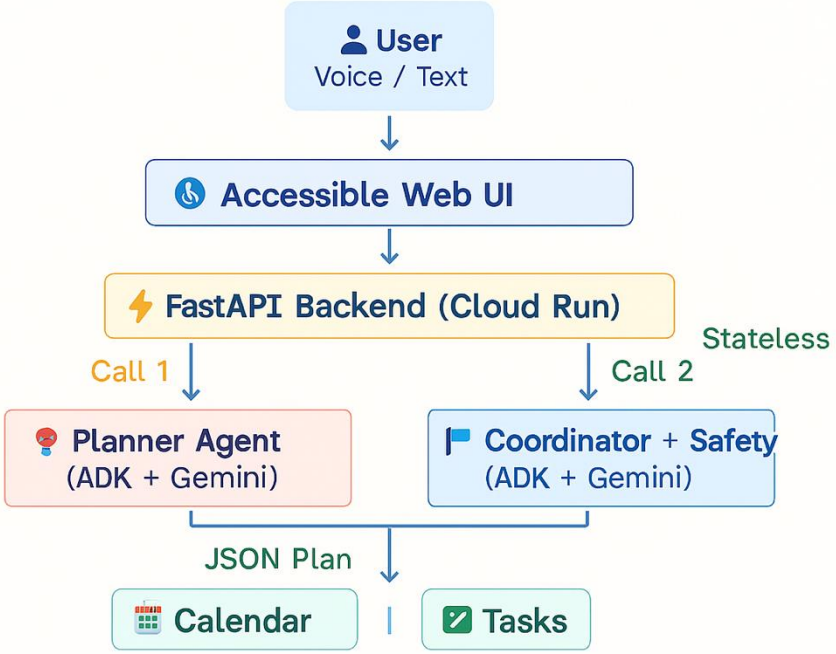
Wireframes / Mock Diagrams

UI Layout — Accessible, Tabbed, Voice-First





Architecture Diagram



Runtime: Cloud Run • Model: Gemini 3.I Flash-Lite
State: In-memory / Firestore

Technologies & Google Services Used

Gemini 3.1 Flash-Lite (Preview)
Most cost-efficient model. Function calling + thinking support. 2-call pipeline within free-tier.

Google Agent Development Kit (ADK)
Native multi-agent orchestration. Each agent has clear role, model config & instructions.

Model Context Protocol (MCP)
Standardized tool interfaces for Calendar and Task Store. Swappable backends.

Google Cloud Run
Serverless container runtime, auto-scaling 0→3 instances. Pay-per-use. Live now.

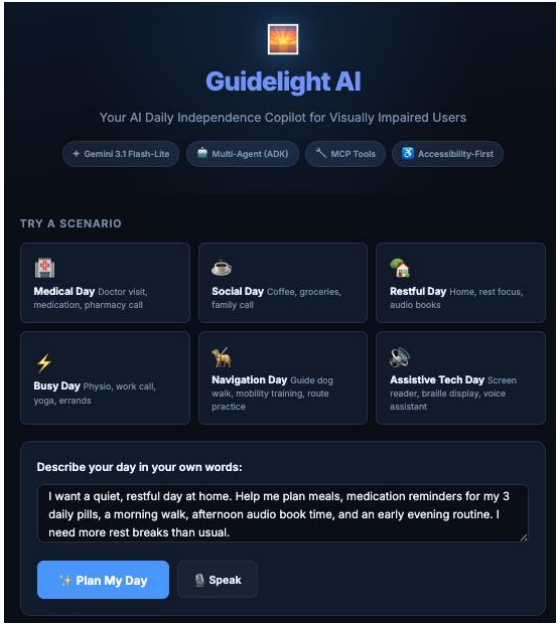
FastAPI (Python 3.11)
Async backend, 5 REST endpoints. Deterministic pipeline for demo reliability.

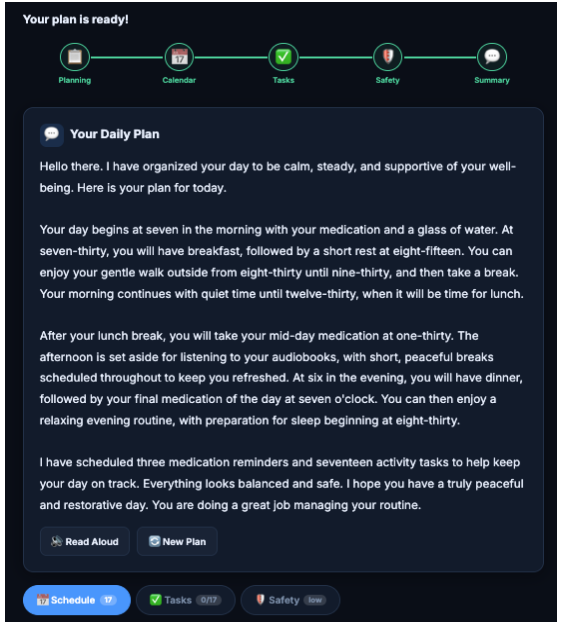
Web Speech API + SpeechSynthesis
Browser-native voice I/O. No third-party deps. Fully hands-free operation.

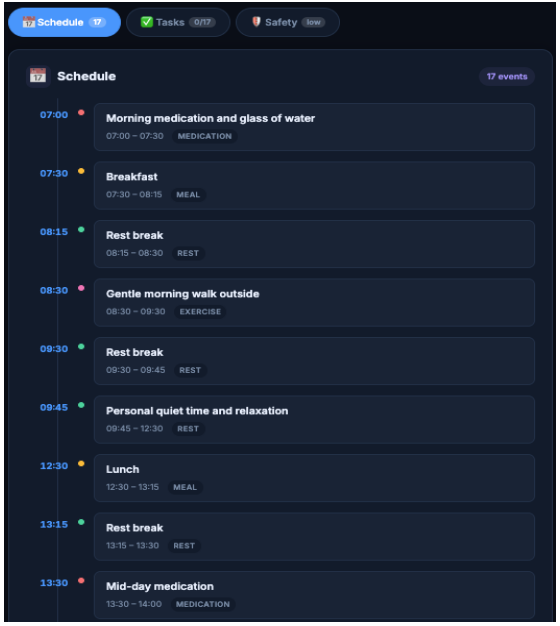
Docker + Cloud Build + Artifact Registry
Containerized. Cloud Build compiles; Artifact Registry stores. Reproducible.

Why this stack? → Optimized for demo reliability, rate-limit survival, accessibility, and real-world deployment — all on free-tier resources.

Snapshots of the Prototype







Live: https://guidelight-ai-669449334512.us-central1.run.app

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Thank you